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Relatório de SGRAI

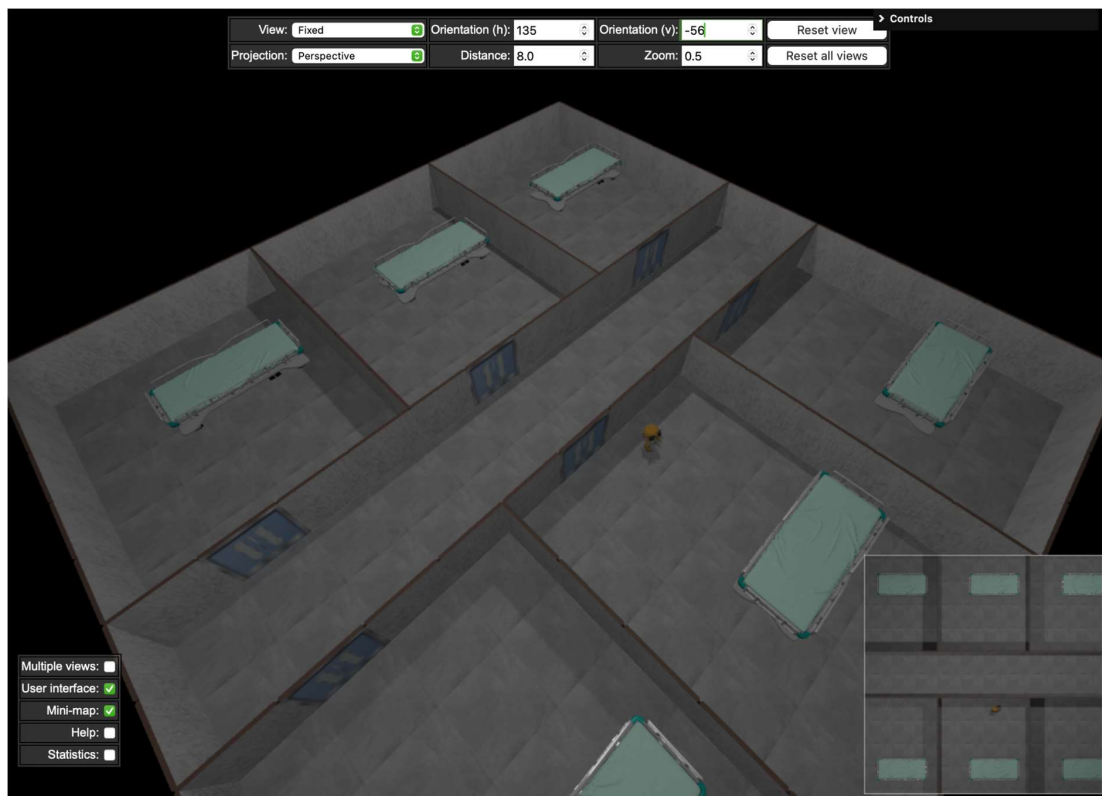
G031 – 3DF

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1. Description of the Project

This project features a 3D visualization of a **hospital floor** plan designed to help healthcare professionals understand and interact with the layout. The representation is dynamically generated from a JSON file and includes **surgical rooms** with key elements such as walls, doors, and surgical tables. Additionally, the app offers enhanced user interactions, such as selecting rooms, controlling the camera view, and opening doors.



2. Technologies Used

- **3D Visualization:** Utilizes Three.js for constructing and managing the 3D environment.
- **Hospital Floor Format:** Employs JSON (JavaScript Object Notation) to define the floor layout and room specifications.
- **Backend Integration:** Interfaces with a backend through HTTP requests to retrieve updated room data.

- **Lighting:** Implements ambient and directional lighting to improve realism and usability.
- **Camera Controls:** Incorporates the OrbitControls addon for intuitive camera navigation.

3. User Interaction

- **Right Mouse Button:** Rotate the view around the scene.
- **Mouse Wheel:** Adjust the zoom level to move closer or farther.
- **Left Mouse Button:** Select a surgical room by clicking on its surgical bed, centering the camera on the chosen room with a smooth transition.
- **“I” Key:** Toggle a pop-up showing room details.
- **“y” Key:** Open Surgery Room door
- **“N” Key:** Close Surgery Room door

4. Conclusion

This assignment achieves the objectives defined for Sprints 2 and 3, offering an interactive and realistic 3D visualization tool for healthcare professionals. It merges user-friendly design with technical reliability, making it well-suited for integration into larger healthcare facilities.